

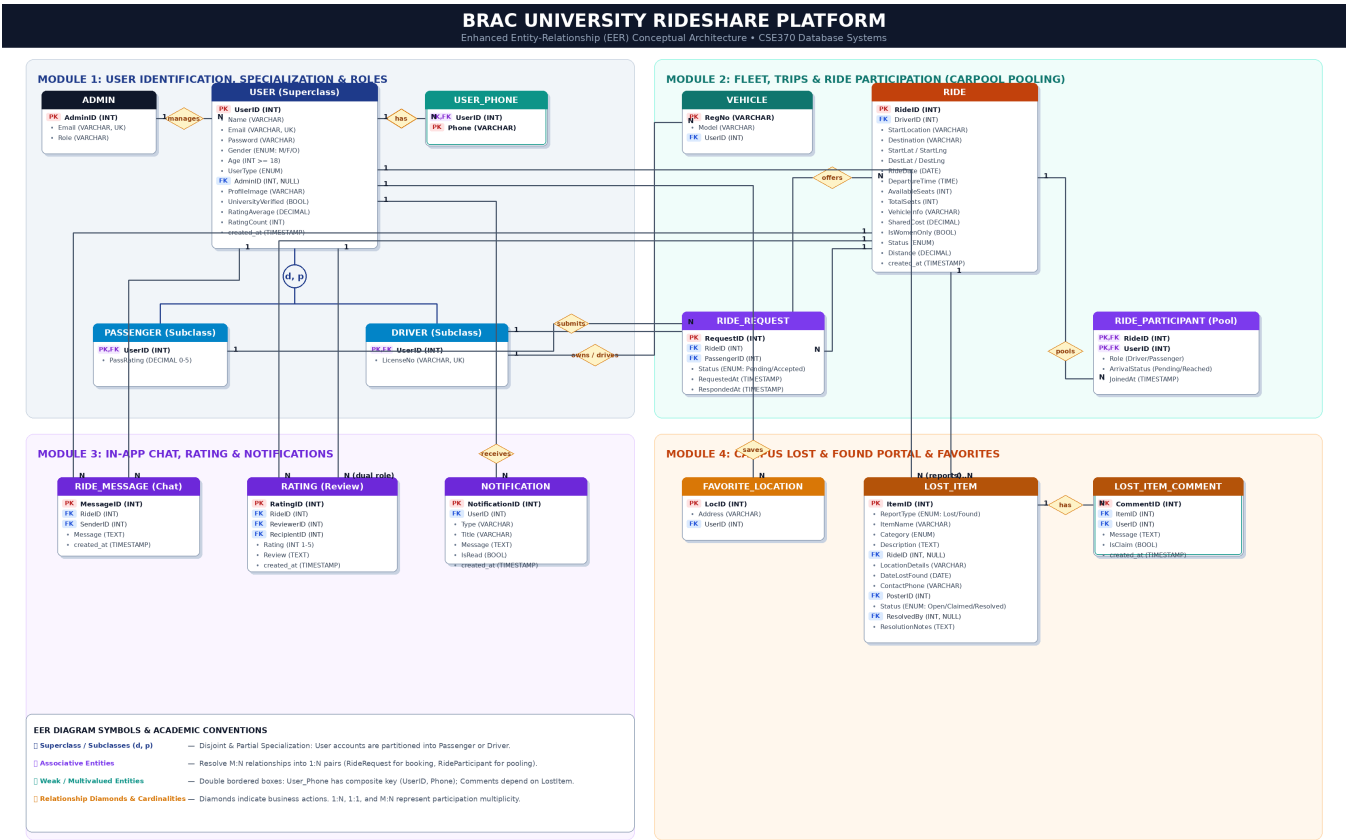
BRAC University Rideshare Platform

Enhanced Entity-Relationship (EER) Model & Relational Schema Specification

Course: CSE370 Database Systems	Database Engine: MySQL 8.0 (InnoDB)
Project: BRACU Rideshare Web Application	Normalization: 3NF / BCNF Compliant
Timezone: Asia/Dhaka (+06:00)	Integrity: Full Referential Cascade Rules

1. Enhanced Entity-Relationship (EER) Conceptual Architecture

The diagram below organizes the BRACU Rideshare system into 4 distinct functional modules: (1) User Identity & Specialization Hierarchy, (2) Fleet & Carpool Pooling, (3) Chat, Rating & Alerts, and (4) Lost & Found Portal.



2. Relationship Cardinality & EER Constraints Matrix

Relationship Name	Entities Involved	Cardinality	Business Meaning & Operational Rule
manages	Admin $\rightarrow$ User	1 : N	Administrators oversee user accounts and university domain verification status.
has_phone	User $\rightarrow$ User_Phone	1 : N	Multivalued contact phone numbers stored without violating 1NF.
specializes_to	User $\rightarrow$ Passenger / Driver	1 : 1 (d, p)	Disjoint & partial inheritance; Drivers store LicenseNo; Passengers store PassRating.
owns_vehicle	Driver $\rightarrow$ Vehicle	1 : N	Drivers register one or more vehicles (Cars, Bikes, CNGs) with unique RegNo.
offers_ride	Driver $\rightarrow$ Ride	1 : N	Drivers publish departure times, available seats, route pins, and Women-Only flags.
ride_request	Passenger $\rightarrow$ Ride	M : N (Associative)	Passengers request seats; Drivers accept/reject via RideRequest entity.
pool_membership	Ride $\rightarrow$ User	M : N (Associative)	RideParticipant tracks confirmed carpool members and live arrival check-in.
in_app_chat	User & Ride $\rightarrow$ RideMessage	1 : N	Active ride participants exchange real-time coordination messages.
review_rating	Ride, Reviewer, Recipient $\rightarrow$ Rating	Ternary (1:N)	Mutual peer feedback ratings (1-5 stars) dynamically updating rolling averages.
lost_and_found	User $\rightarrow$ LostItem $\rightarrow$ Comments	1 : N	Campus-wide and ride-specific item reporting, claiming, and resolution pipeline.

3. Complete Relational Database Schema & Data Dictionary

Formal database catalog specification detailing data types, keys, constraints, and cascade actions:

Table: Admin — System administrators overseeing verifications and platform moderation

AdminID	INT	PK	AUTO_INC	Primary Key
Email	VARCHAR(100)	UK	NOT NULL	Unique administrator email
Role	VARCHAR(50)	-	NOT NULL	e.g. 'SuperAdmin', 'Moderator'

Table: User — Superclass account entity for all campus members (Passenger, Driver, Admin)

UserID	INT	PK	AUTO_INC	Primary Key
Name	VARCHAR(100)	-	NOT NULL	Full Name
Email	VARCHAR(100)	UK	NOT NULL	Unique University/Personal Email
Password	VARCHAR(255)	-	NOT NULL	Bcrypt Hashed Password
Gender	ENUM	-	NOT NULL	ENUM('Male', 'Female', 'Other')
Age	INT	-	NULL	CHECK (`Age` >= 18)
UserType	ENUM	-	DEFAULT 'Passenger'	ENUM('Passenger', 'Driver', 'Admin')
AdminID	INT	FK	NULL	FK -> Admin(AdminID) ON DELETE SET NULL
ProfileImage	VARCHAR(255)	-	NULL	Relative avatar image path
UniversityVerified	TINYINT(1)	-	DEFAULT 0	1 if verified BRACU student/faculty
RatingAverage	DECIMAL(3,2)	-	DEFAULT 5.00	Rolling average reputation score
RatingCount	INT	-	DEFAULT 0	Total feedback reviews count
created_at	TIMESTAMP	-	CURRENT_TIMESTAMP	Account creation timestamp

Table: User_Phone — Multivalued phone attribute stored in 1NF normalized relation

UserID	INT	PK, FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
Phone	VARCHAR(20)	PK	NOT NULL	Contact phone number

Table: Passenger — Specialization subclass for passenger accounts

UserID	INT	PK, FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
PassRating	DECIMAL(3,2)	-	DEFAULT 5.00	CHECK (`PassRating` BETWEEN 0.00 AND 5.00)

Table: Driver — Specialization subclass for verified campus drivers

UserID	INT	PK, FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
LicenseNo	VARCHAR(50)	UK	NOT NULL	BRTA Driving License Number

Table: Vehicle — Registered vehicles owned and operated by verified drivers

RegNo	VARCHAR(50)	PK	NOT NULL	Vehicle Registration Number (BRTA)
Model	VARCHAR(100)	-	NOT NULL	Model/Brand (e.g. Toyota Axio, Honda CB)
UserID	INT	FK	NOT NULL	FK -> Driver(UserID) ON DELETE CASCADE

Table: FavoriteLocation — Saved addresses for rapid single-click booking and search

LocID	INT	PK	AUTO_INC	Primary Key
Address	VARCHAR(255)	-	NOT NULL	Saved pickup/destination string
UserID	INT	FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE

Table: Ride — Core entity representing offered carpool trips across Dhaka

RideID	INT	PK	AUTO_INC	Primary Key
DriverID	INT	FK	NOT NULL	FK -> Driver(UserID) ON DELETE CASCADE
StartLocation	VARCHAR(255)	-	NOT NULL	Starting pickup area name
Destination	VARCHAR(255)	-	NOT NULL	Drop-off destination area name
StartLat / Lng	DECIMAL(10,8)	-	NULL	GPS latitude and longitude coordinates
DestLat / Lng	DECIMAL(10,8)	-	NULL	GPS latitude and longitude coordinates
RideDate	DATE	-	NOT NULL	Date of scheduled trip
DepartureTime	TIME	-	NOT NULL	Planned departure time
AvailableSeats	INT	-	NOT NULL	Available seats remaining
TotalSeats	INT	-	NOT NULL	Total passenger capacity
VehicleInfo	VARCHAR(255)	-	NULL	Vehicle description / type
SharedCost	DECIMAL(10,2)	-	DEFAULT 0.00	Fuel share per passenger (BDT)
IsWomenOnly	TINYINT(1)	-	DEFAULT 0	1 = Restricted to female passengers only
Status	ENUM	-	DEFAULT 'Open'	ENUM('Open', 'Full', 'Confirmed', 'Completed', ...)
Distance	DECIMAL(6,2)	-	NULL	Estimated route distance in km

Table: RideRequest — Passenger booking applications and driver decisions (M:N resolver)

RequestID	INT	PK	AUTO_INC	Primary Key
RideID	INT	FK	NOT NULL	FK -> Ride(RideID) ON DELETE CASCADE
PassengerID	INT	FK	NOT NULL	FK -> Passenger(UserID) ON DELETE CASCADE
Status	ENUM	-	DEFAULT 'Pending'	ENUM('Pending', 'Accepted', 'Rejected')
RequestedAt	TIMESTAMP	-	CURRENT_TIMESTAMP	Request submission time
RespondedAt	TIMESTAMP	-	NULL	Driver decision timestamp

Table: RideParticipant — Associative entity managing confirmed carpool pooling & arrivals

RideID	INT	PK, FK	NOT NULL	FK -> Ride(RideID) ON DELETE CASCADE
UserID	INT	PK, FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
Role	ENUM	-	DEFAULT 'Passenger'	ENUM('Driver', 'Passenger')
ArrivalStatus	ENUM	-	DEFAULT 'Pending'	ENUM('Pending', 'Reached', 'Not Reached')
JoinedAt	TIMESTAMP	-	CURRENT_TIMESTAMP	Confirmation timestamp

Table: Rating — Peer review system evaluating rides and computing user ratings

RatingID	INT	PK	AUTO_INC	Primary Key
RideID	INT	FK	NOT NULL	FK -> Ride(RideID) ON DELETE CASCADE
ReviewerID	INT	FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
RecipientID	INT	FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
Rating	INT	-	NOT NULL	CHECK ('Rating' BETWEEN 1 AND 5)
Review	TEXT	-	NULL	Written feedback comment
created_at	TIMESTAMP	-	CURRENT_TIMESTAMP	Review date

Table: RideMessage — In-app active ride group chat and coordination messages

MessageID	INT	PK	AUTO_INC	Primary Key
RideID	INT	FK	NOT NULL	FK -> Ride(RideID) ON DELETE CASCADE
SenderID	INT	FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
Message	TEXT	-	NOT NULL	Chat message body
created_at	TIMESTAMP	-	CURRENT_TIMESTAMP	Message timestamp

Table: LostItem — Campus and in-vehicle lost and found reporting portal

ItemID	INT	PK	AUTO_INC	Primary Key
ReportType	ENUM	-	NOT NULL	ENUM('Lost', 'Found')
ItemName	VARCHAR(150)	-	NOT NULL	Item title
Category	ENUM	-	DEFAULT 'Other'	ENUM('Electronics', 'Student ID', 'Keys', ...)
Description	TEXT	-	NOT NULL	Detailed characteristics
RideID	INT	FK	NULL	FK -> Ride(RideID) ON DELETE SET NULL
PosterID	INT	FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
Status	ENUM	-	DEFAULT 'Open'	ENUM('Open', 'Claimed', 'Resolved')
ResolvedBy	INT	FK	NULL	FK -> User(UserID) ON DELETE SET NULL

Table: LostItemComment — Community claim inquiries and recovery comments on lost items

CommentID	INT	PK	AUTO_INC	Primary Key
ItemID	INT	FK	NOT NULL	FK -> LostItem(ItemID) ON DELETE CASCADE
UserID	INT	FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
Message	TEXT	-	NOT NULL	Comment or proof of ownership
IsClaim	TINYINT(1)	-	DEFAULT 0	1 = Official ownership claim
created_at	TIMESTAMP	-	CURRENT_TIMEST AMP	Comment timestamp

Table: Notification — System alerts for ride requests, chat updates, and approvals

NotificationID	INT	PK	AUTO_INC	Primary Key
UserID	INT	FK	NOT NULL	FK -> User(UserID) ON DELETE CASCADE
Type	VARCHAR(50)	-	NOT NULL	e.g. 'ride_request', 'chat', 'review'
Title	VARCHAR(100)	-	NOT NULL	Notification title
Message	TEXT	-	NOT NULL	Notification body
IsRead	TINYINT(1)	-	DEFAULT 0	0 = Unread, 1 = Read
Link	VARCHAR(255)	-	NULL	Direct redirect URL

4. Academic Integrity & Normalization Compliance

1. Third Normal Form (3NF) & Boyce-Codd Normal Form (BCNF): Every non-prime attribute is fully functionally dependent on the primary key, and no transitive dependencies exist across non-prime attributes. Multivalued phone numbers are normalized into User_Phone.

2. Specialization Constraints: Generalization of User to Passenger and Driver adheres to a **Disjoint (d)** and **Partial (p)** constraint rule, allowing extensible role-specific attributes while preventing duplicate identity records.

3. Referential Integrity & Cascade Semantics: All relationships enforce strict cascading deletes on operational children to eliminate dangling pointers, while preserving historical audit logs using ON DELETE SET NULL for optional references.